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COLLEGE OF ENGINEERING & MANAGEMENT
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MANGALURU

Major Project Phase-1 Presentation

Explainable Federated Learning Framework for Privacy-Preserving Healthcare Systems

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Introduction

Artificial Intelligence (AI) is becoming an important tool in healthcare, helping doctors predict diseases and make better clinical decisions. However, most traditional AI systems depend on centralized data storage, which can create privacy and security concerns for sensitive patient information. Strict regulations also make it difficult for hospitals to share data with each other. While Federated Learning allows different institutions to train AI models together without sharing raw data, there is still a need for secure, scalable, and transparent platforms to support effective collaboration in healthcare AI.

Problem Statement

Healthcare organizations lack a secure, privacy-preserving, and transparent platform to collaboratively develop AI models while protecting patient data and ensuring trust in clinical decisions.

Literature Review

SL. NO.	Author	Title of the paper	Methodology	Key Findings/Limitations
1	Misbah Bibi et al. (2026)	Explainable Multi-Modal Fusion-Based Federated Learning for Mortality Prediction in Energy-Constrained Healthcare Systems	Hybrid transformer (clinical notes) + tree-based model (structured data) within FL; evaluated on MIMIC-IV dataset.	Achieved AUC 0.89 for mortality prediction, providing interpretable insights via multimodal fusion. Limitation: communication overhead and scalability challenges in multimodal FL.
2	Florenc Demrozi et al. (2026)	Multimodal AI for Next-Generation Healthcare: Data Domains, Algorithms, Challenges, and Future Perspectives	Opinion review of multimodal AI (MMAI) integrating imaging, EHR, genomics, sensors, social/behavioral data.	MMAI enables predictive, personalized, and preventive healthcare using fusion strategies like early, intermediate, late, contrastive, and hybrid. Limitations include interoperability, explainability, regulatory constraints, and high computational demands.

3	Rahmati et al. (2025)	Federated Learning and Explainable AI for Personalized Healthcare in Resource-Limited Settings	Lightweight federated learning framework with SHAP/LIME explainability, optimized for low-resource environments and multi-modal data.	Improved predictive accuracy (F1-score 0.85), reduced communication cost by 40% and inference time by 30%, with a fidelity score of 0.92 for explanations. Limitation: requires robustness against adversarial attacks and enhanced explainability for complex modalities.
4	Tanisha Bhardwaj & K. Sumangali (2025)	An Explainable Federated Blockchain Framework with Privacy-Preserving AI Optimization for Securing Healthcare Data	PPFBXAIO framework combining blockchain (SHA-256), Explainable AI (SHAP), and Levy Grasshopper Optimization Algorithm (LGOA) for feature selection.	Enhanced accuracy (≈93–95%) on heart disease and breast cancer datasets, with reduced latency and improved throughput compared to FedAvg and PEFL. Limitation: optimization and explainability integration increases computational cost.
5	Rahul Haripriya et al. (2025)	Privacy-Preserving Federated Learning for Collaborative Medical Data Mining in Multi-Institutional Settings	Adaptive aggregation (FedAvg + FedSGD) with transfer learning (GoogLeNet, VGG16, EfficientNetV2, ResNet-RS).	Dynamic aggregation improves convergence and accuracy, achieving strong performance on TB X-rays, brain tumor MRI, and diabetic retinopathy datasets. Limitation: computational overhead for larger models.

6	Tanisha Bhardwaj & K. Sumangali (2025)	A Federated Incremental Blockchain Framework with Privacy Preserving XAI Optimization for Securing Healthcare Data	Proposed PPFILB-OXAI framework integrating Blockchain, Federated Incremental Learning (FIL), Explainable AI, and Chaotic Bobcat Optimization Algorithm (CBOA).	FIL adapts to new data without retraining, blockchain ensures transparency and security, and CBOA improves feature selection, achieving ≈95–97% accuracy on heart disease and breast cancer datasets. Limitation: increased complexity and potential deployment latency.
7	Tanjila Alam Sathi et al. (2025)	Explainable Federated Learning for Multi-Class Heart Disease Diagnosis via ECG Fiducial Features	FL-LSTM model trained on MIT-BIH, European ST-T, and FANTASIA datasets; integrated SHAP and LIME for interpretability.	Achieved 92% accuracy, 99% AUC, and 91% F1 score, providing global and local interpretability of ECG features. Limitation: computationally intensive and requires handling class imbalance with SMOTE.
8	D. Cenitta et al. (2025)	Explainable AI With Homomorphic Encryption for Secure Cloud-Based ECG Analysis in Heart Disease Diagnosis	CNN-based diagnostic framework using CKKS homomorphic encryption and SHAP explanations on MIT-BIH dataset.	Achieved 94.2% accuracy (plaintext) and 93.4% (encrypted), with encrypted inference latency ≈420 ms and clinician-validated SHAP explanations. Limitation: computational overhead of homomorphic encryption.

9	Sayed Tariq Shah et al. (2025)	Federated Learning in Public Health: A Systematic Review of Decentralized, Equitable, and Secure Disease Prevention Approaches	PRISMA-based systematic review of 19 studies (2020–2025).	FL is applied to communicable diseases (COVID-19, TB) and chronic conditions, offering privacy preservation, generalizability, and near real-time intelligence. Limitations include non-IID data, interoperability gaps, and limited deployment in low-resource settings.
10	Hagyeong Ryu et al. (2025)	Federated Learning for Cardiovascular Disease Prediction: Comparative Review of Biosignal- and EHR-Based Approaches	Review of 28 studies (2020–2025), comparing biosignal (ECG, PPG, PCG) vs. EHR-based FL.	Biosignal FL enables personalization, IoT monitoring, and handling non-IID data, while EHR FL supports large-scale collaboration, adaptive optimization, and secure aggregation. Limitation previous disease-centric reviews lacked modality-specific insights; this work offers design guidelines.
11	Sileshi Nibret Zeleke & Mario Bochicchio (2024)	Towards Explainable Federated Learning in Healthcare: Heart Arrhythmia Detection	Attention-based Temporal Convolutional Network (TCN) with FL on MIT-BIH dataset.	Achieved balanced accuracy 93.5% and F1-score 98.7%, with attention weights enhancing ECG interpretability. Limitations include small dataset, communication overhead, and non-IID data challenges.

12	Liutao Zhao et al. (2024)	Explainable Federated Learning Scheme for Secure Healthcare Data Sharing	Proposed a federated learning scheme with post-hoc explainability and a multi-server secure aggregation protocol using homomorphic secret sharing.	Uses gradient contribution analysis for explainability and a multi-server architecture to prevent single-point failures, ensuring verifiable secure aggregation even with colluding servers. Limitations include partial integration of explanations into training and high computational overhead.
13	Sarthak Pati et al. (2024)	Preservation for Federated Learning in Healthcare	Review of privacy threats and mitigation strategies (CIA triad framework).	FL faces risks like model inversion, membership inference, and poisoning attacks. Mitigation strategies include differential privacy, secure multi-party computation, and encryption, but challenges remain due to privacy–performance trade-offs and overhead costs.
14	Zhen Ling Teo et al. (2024)	Federated Machine Learning in Healthcare: A Systematic Review on Clinical Applications and Technical Architecture	Systematic review of 612 studies (2018–2023).	Radiology and internal medicine dominate FL applications, but only 5.2% of studies report real-world deployment. Privacy mechanisms include differential privacy, homomorphic encryption, and blockchain, while key limitations remain lack of standardized reporting and limited clinical translation.

15	Fayez Saud Alreshidi et al. (2024)	Fed-CL: An Atrial Fibrillation Prediction System Using ECG Signals Employing Federated Learning Mechanism	Hybrid CNN-LSTM federated learning model for AFib prediction using ECG signals and heart rate variability features.	Achieved high accuracy (up to 99.4% on MIT-BIH AF database) while preserving privacy by keeping patient data local. Limitations include dataset imbalance, heterogeneity affecting generalization, and limited interpretability of deep models.
16	Li et al. (2024)	From Challenges and Pitfalls to Recommendations and Opportunities in Federated Learning for Healthcare	Conceptual review (blocked content, but metadata indicates it's a critical discussion piece).	Highlights pitfalls in FL deployment in healthcare and offers recommendations to overcome technical, ethical, and governance barriers. Limitation: conceptual scope with no full empirical data available.
17	Koutsoubis et al. (2024)	Privacy Preserving Federated Learning in Medical Imaging with Uncertainty Estimation	Empirical study (blocked content, but metadata indicates focus on medical imaging).	Combines FL with uncertainty estimation for medical imaging, enhancing trustworthiness by quantifying prediction uncertainty. Limitation: likely computational overhead and complexity in uncertainty quantification.
18	Md Saifur Rahman et al. (2024)	Application of Federated Learning in Cardiology: Key Challenges and Potential Solutions	Commentary on federated learning applications in cardiology, analyzing IoT-based data sharing and challenges across layers (application, transmission, cloud).	FL faces challenges like signal quality, attacks, communication cost, latency, malicious clients, and scalability; solutions include encryption, differential privacy, edge computing, and anomaly detection, though broader validation is needed.

19	Ahmad Chaddad et al. (2022)	Explainable, Domain-Adaptive, and Federated Artificial Intelligence in Medicine	Narrative review of three methodological approaches —Explainable AI (XAI), Domain Adaptation (DA), and Federated Learning (FL) —with emphasis on medical applications.	XAI improves model transparency but only partially explains complex models, while Domain Adaptation (DA) addresses domain shift in heterogeneous medical data. Federated Learning (FL) enables collaborative training without sharing raw data but faces privacy/security challenges, and integration of XAI, DA, and FL for clinical practice remains limited.
20	Nicola Rieke et al. (2020)	The Future of Digital Health with Federated Learning	Perspective paper outlining FL's potential in digital health, comparing centralized vs. federated workflows.	Federated Learning (FL) supports privacy-preserving collaboration across institutions, with early use in MRI, mammography, and tumor segmentation. Key limitations include technical heterogeneity, privacy leakage risks, and lack of standards.

Objectives

Develop a Multimodal Healthcare Prediction Framework

- To design and implement a multimodal disease prediction system using medical images, clinical tabular data, and voice signals to improve disease detection accuracy.

Integrate Explainable Artificial Intelligence (XAI)

- To integrate explainable AI techniques such as SHAP, Grad-CAM, and spectrogram attention to provide transparent and interpretable healthcare predictions.

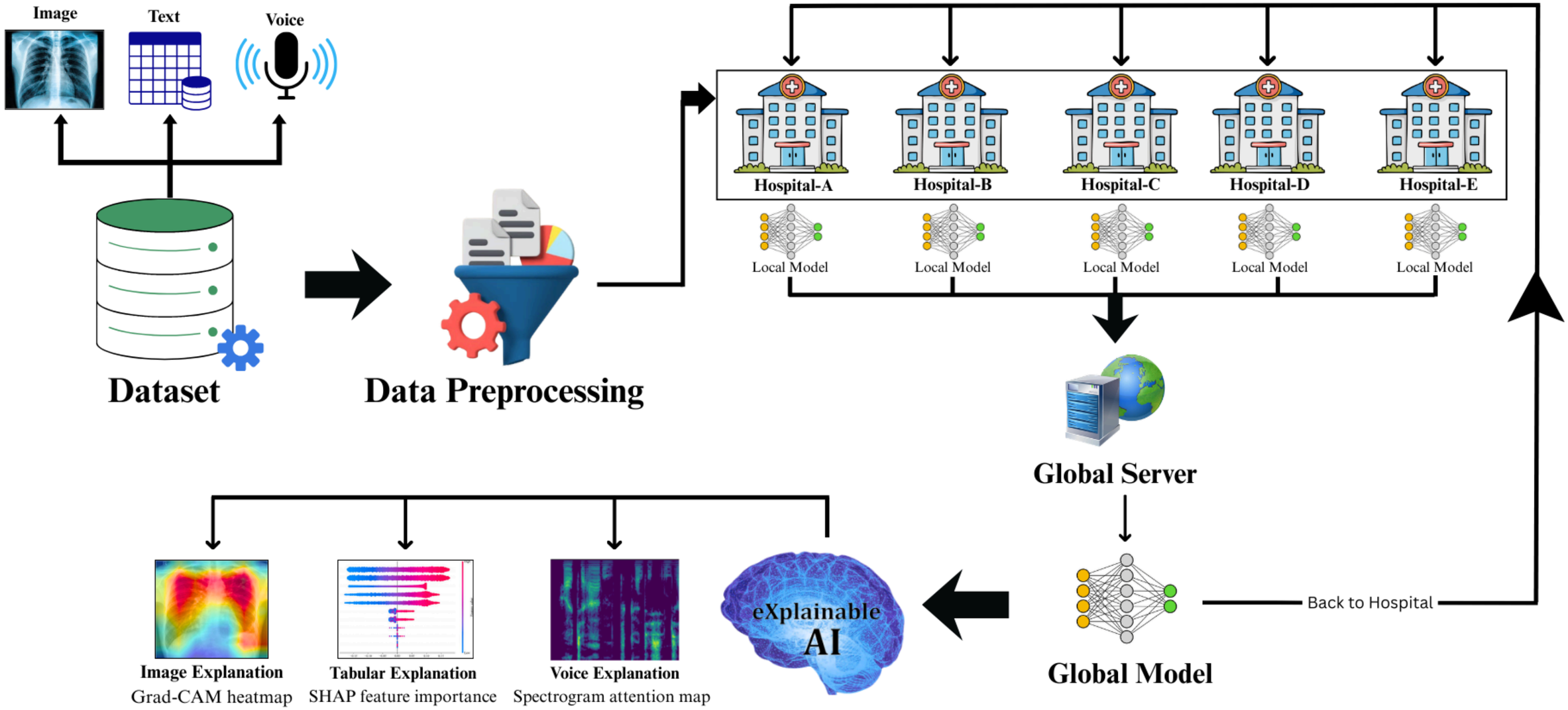
Implement Privacy-Preserving Federated Learning

- To develop a federated learning system that enables hospitals to collaboratively train models while preserving patient data privacy using the FedAvg algorithm.

Evaluate and Compare Model Performance

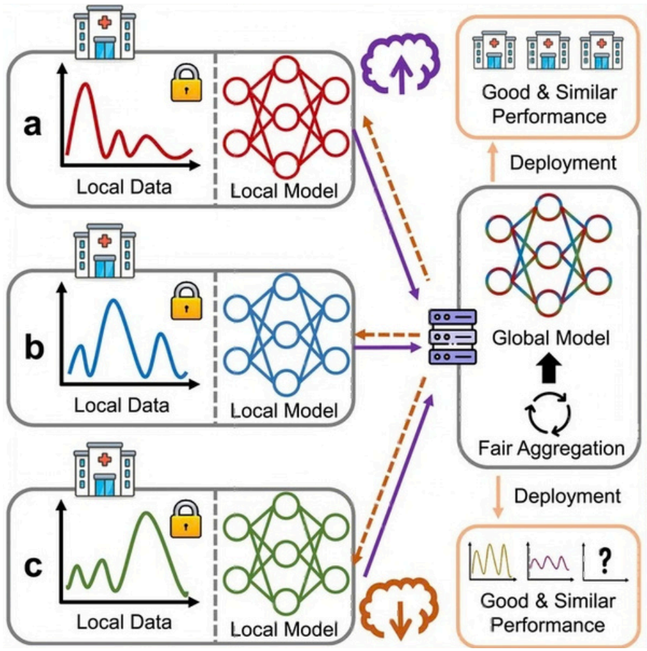
- To evaluate and compare machine learning models across multiple data modalities using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

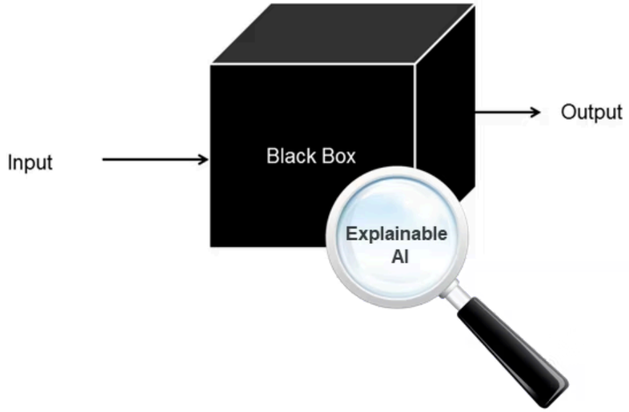
System Design/ Architecture



Expected Outcome

1. Improved Disease Prediction Accuracy
2. Privacy-Preserving Collaborative Learning System
3. Comparative Analysis of Machine Learning Models
4. Interpretable Healthcare Predictions





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Thank You